

1 solving congruences

imagine it is 10 o'clock. 5 hours later, it is 3 o'clock, not 15. this "resetting" behaviour is the essence of **modular arithmetic**.

1.1 arithmetic operations with $\text{mod } n$

$\mathbb{Z}_m$ denotes the set of non-negative integers less than m (the remainders when dividing by m).

modular arithmetic "preserves" addition and multiplication:

$$(a + b) \bmod m = ((a \bmod m) + (b \bmod m)) \bmod m$$

$$(a \cdot b) \bmod m = ((a \bmod m)(b \bmod m)) \bmod m$$

we define operations restricted to this set:

$$a +_m b = (a + b) \bmod m$$

$$a \cdot_m b = (a \cdot b) \bmod m$$

1.2 solving linear congruences

we look at equations of the form $ax \equiv b \pmod{m}$.

1.2.1 additive inverses

look at an example: in order to solve $a + 8 \equiv 2 \pmod{11}$:

1. find the **additive inverse** of 8 modulo 11.

- $-8 \equiv 3 \pmod{11}$
- $-8 = (11 \cdot -1) + 3$

2. add 3 to both sides: $a \equiv 5 \pmod{11}$.

note: adding any multiple of m preserves the value $\pmod{m}$.

1.2.2 multiplicative inverses

to solve $ax \equiv b \pmod{m}$ (a **linear congruence**), given values for a and b , we need a value $\bar{a}$ such that $a \cdot \bar{a} \equiv 1 \pmod{m}$. if we have this value, we can multiply on both sides, and then simplify. **bézout's theorem** helps to find that number.

theorem (bézout's theorem): if $\gcd(a, m) = 1$ (a and m are coprime), there exist integers s and t such that: $sa + tm = 1$.

this implies $sa \equiv 1 \pmod{m}$, meaning s is the multiplicative inverse of a .

collary: since $\gcd(a, m) = 1$ and $sa + tm = 1$, by substitution, we can get the **bézout identity**: $\gcd(a, b) = sa + tb$.

1.3 extended euclidean algorithm (eea)

we utilise the **extended euclidean algorithm** to find the gcd and the bézout numbers (s, t) .

example: find gcd and bézout numbers for 99 and 78. we track $r_i = r_{i-2} - q_i r_{i-1}$ and work backwards (or track s, t alongside):

row	a	b	formula
1	99	78	$99 = 1 \cdot 78 + 21$
2	78	21	$78 = 3 \cdot 21 + 15$
3	21	15	$21 = 1 \cdot 15 + 6$
4	15	6	$15 = 2 \cdot 6 + 3$
5	6	3	$6 = 2 \cdot 3 + 0$

$\gcd(99, 78) = 3$. back-substitution yields $s = -11, t = 14$, so:

$$99(-11) + 78(14) = 3$$

application: solve the linear congruence $57x \equiv 5 \pmod{98}$.

1. eea on 98 and 57 gives: $98(-25) + 57(43) = 1$.
2. inverse of 57 mod 98 is 43.
3. multiply both sides by 43:

$$x \equiv 5 \cdot 43 \equiv 215 \equiv 19 \pmod{98}$$

1.4 chinese remainder theorem (crt)

used to solve systems of congruences where moduli are pairwise coprime.

assume we have a system like this:

$$x \equiv a_1 \pmod{m_1}$$

$$x \equiv a_2 \pmod{m_2}$$

...

$$x \equiv a_n \pmod{m_n}$$

solution: let $m = m_1 m_2 \dots m_n$. let $M_k = \frac{m}{m_k}$ (product of all moduli except m_k). let y_k be the inverse of M_k modulo m_k .

the unique solution modulo m is:

$$x = \sum_{k=1}^n a_k M_k y_k$$

example (sunzi suanjing):

$$x \equiv 2 \pmod{3}, x \equiv 3 \pmod{5}, x \equiv 2 \pmod{7}$$

- $m = 3 \cdot 5 \cdot 7 = 105$.
- $M_1 = 35, y_1 = 2$ (since $35 \cdot 2 = 70 \equiv 1 \pmod{3}$).
- $M_2 = 21, y_2 = 1$ (since $21 \cdot 1 = 21 \equiv 1 \pmod{5}$).
- $M_3 = 15, y_3 = 1$ (since $15 \cdot 1 = 15 \equiv 1 \pmod{7}$).
- $x = 2(35)(2) + 3(21)(1) + 2(15)(1) = 140 + 63 + 30 = 233 \equiv 23 \pmod{105}$.

1.5 advanced concepts

1.5.1 fermat's little theorem

if p is prime and p does not divide a :

$$a^{p-1} \equiv 1 \pmod{p}$$

useful for computing large powers, e.g., $7^{222} \pmod{11}$.

1.5.2 primitive roots & discrete log

- **primitive root (r):** an element in $\mathbb{Z}_p$ whose powers generate all nonzero elements of $\mathbb{Z}_p$.
- **discrete log problem:** given p, r, b , find e such that $r^e \equiv b \pmod{p}$. there is no known polynomial time algorithm for this (foundation of cryptography).